

Akash Reddy Kukkala

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Education

University of Maryland, College Park, US

Jan 2025 – Dec 2026

Master of Engineering, Cybersecurity Engineering

GPA: 3.9/4.0

Jawaharlal Nehru Technological University, Telangana, India

Sep 2020 – May 2024

Bachelor of Technology, Computer Science and Engineering (Cybersecurity)

GPA: 7.5/10

Experience

Penetration Testing Intern, Coincent

September 2022 – November 2023

- Conducted web application and API penetration tests identifying OWASP Top 10 vulnerabilities (SQLi, XSS, IDOR) across client environments using Burp Suite, OWASP ZAP, and Metasploit.
- Performed network reconnaissance and enumeration using Nmap to map attack surfaces and uncover misconfigurations across client infrastructure.
- Authored structured pentest reports translating technical findings into risk-rated remediation guidance; supported incident response efforts by tracking vulnerability lifecycle from discovery through remediation.

Projects

AWS Secure Vault with Threat Detection

In Progress

- Built a serverless file vault on AWS: Cognito (JWT) → API Gateway → Lambda (Python) → S3 with SSE-KMS encryption.
- Deployed CloudTrail, CloudWatch Alarms, and SNS for audit logging and real-time security alerting on all vault access events.
- Integrated GuardDuty for threat detection and deployed frontend via AWS Amplify with Cognito-backed authentication.

AWS Cloud Security Hardening (E-Commerce Infrastructure)

May 2025

- Assessed an AWS-hosted e-commerce platform (OWASP Juice Shop) via CloudFormation, identifying 12+ critical vulnerabilities across IAM, VPC, EBS, and compute layers using AWS CLI and Nmap.
- Hardened EC2 by enforcing least-privilege IAM, restricting SSH to a trusted CIDR, and enabling EBS encryption via KMS.
- Automated EBS snapshots via AWS Backup, deployed CloudWatch Agent for metrics, and configured CloudTrail for API audit logging to S3.

DDoS Attack Detection System

August 2025

- Built a hybrid intrusion detection system on NSL-KDD dataset classifying traffic into 5 categories (Normal, DoS, Probe, R2L, U2R) using Random Forest, SVM, Logistic Regression, and Decision Tree.
- Developed an end-to-end ML pipeline with one-hot encoding, Z-Score normalization, class imbalance handling, and feature selection using Python and Scikit-learn.
- Deployed a Flask web app for real-time attack simulation, letting users input packet features and predict live attack-type.

Research Publications

DDoS Attack Detection using Machine Learning

April 2024

International Journal of Innovative Research in Technology (IJIRT)

- Proposed a hybrid ML-based DDoS detection framework combining ensemble classifiers to accurately identify malicious network traffic and improve resilience against sophisticated attacks.

Keylogger Detection System

December 2024

International Journal of Innovative Research in Technology (IJIRT)

- Analyzed keylogger behavior in credential theft and forensic contexts, examining dual-use implications across offensive and defensive cybersecurity applications.

Certifications & Trainings

AWS Certified Cloud Practitioner

February 2026

eLearn Junior Penetration Tester (eJPT)

September 2025

Ethical Hacking Essentials

November 2024

Microsoft Certified: SC-900

September 2022

CompTIA Security+

In Progress

Certified Defensive Security Analyst (CDSA)

In Progress

Technical Skills

Cloud & AWS: EC2, S3, IAM, VPC, Lambda, API Gateway, CloudTrail, CloudWatch, GuardDuty, Security Hub, KMS, AWS Backup, Config, Cognito, Amplify, SNS, CloudFormation, AWS CLI

Security Tools: Burp Suite, OWASP ZAP, Nmap, Metasploit, Wireshark, Hydra, sqlmap, Splunk, CSPM (AWS Security Hub, Config), John the Ripper, GDB, Pwndbg, Ghidra, Radare2, objdump, strace

Programming: Python, Bash, PowerShell, JavaScript, SQL, C, Java, PHP, HTML/CSS

Infrastructure & DevOps: IaC (Terraform, CloudFormation), Docker, Kubernetes, Git, GitHub Actions, CI/CD